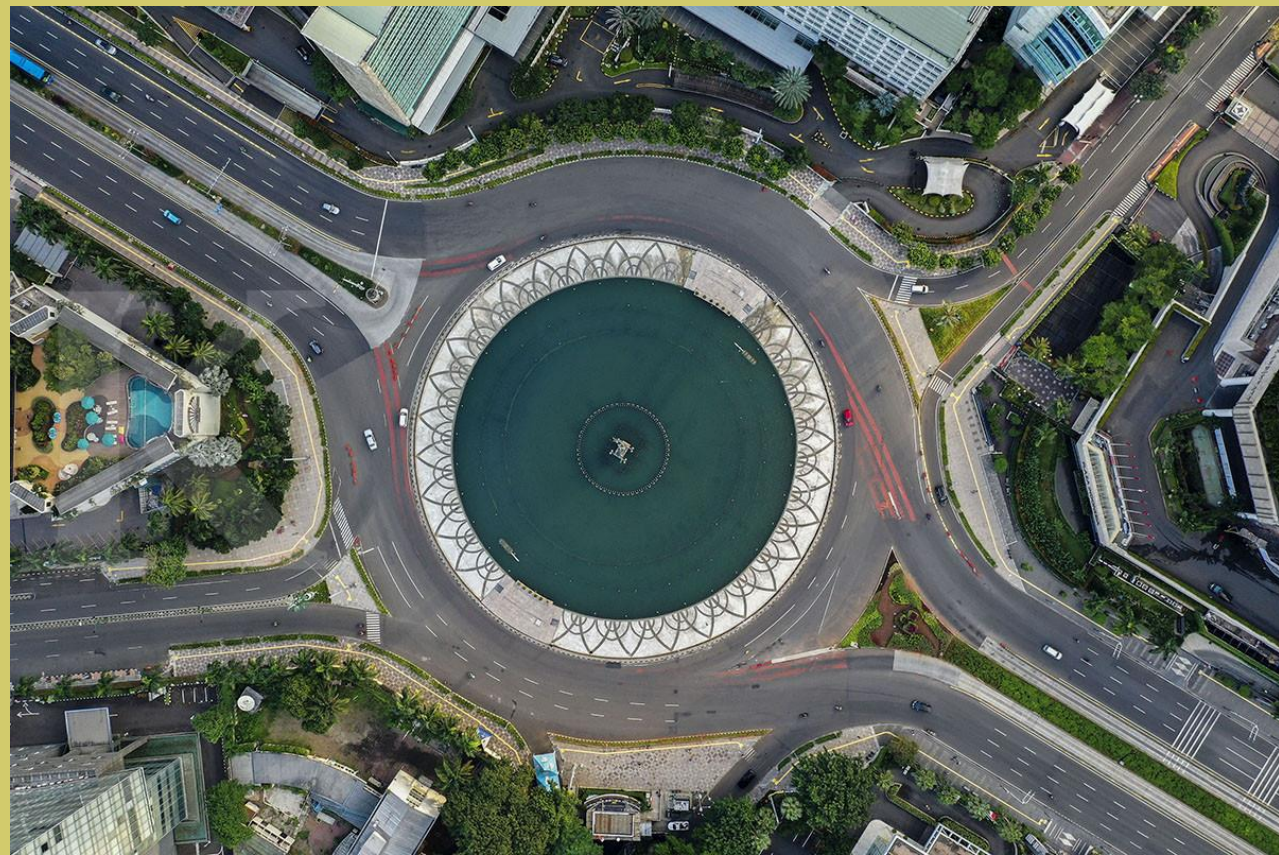




ALGORITMA PEMROGRAMAN - ALPRO

PRODI INFORMATIKA

Enny Itje Sela



PERULANGAN/LOOPING

FOR
WHILE

FOR & WHILE

LOOPING DENGAN FOR

TAHU BERAPA KALI HARUS DIULANG

ADA NILAI AWAL & NILAI AKHIR

LOOPING BERHENTI JIKA NILAI AKHIR
DICAPAI

LOOPING DENGAN WHILE

TIDAK DIKETAHUI BERAPA KALI HARUS
DIULANG

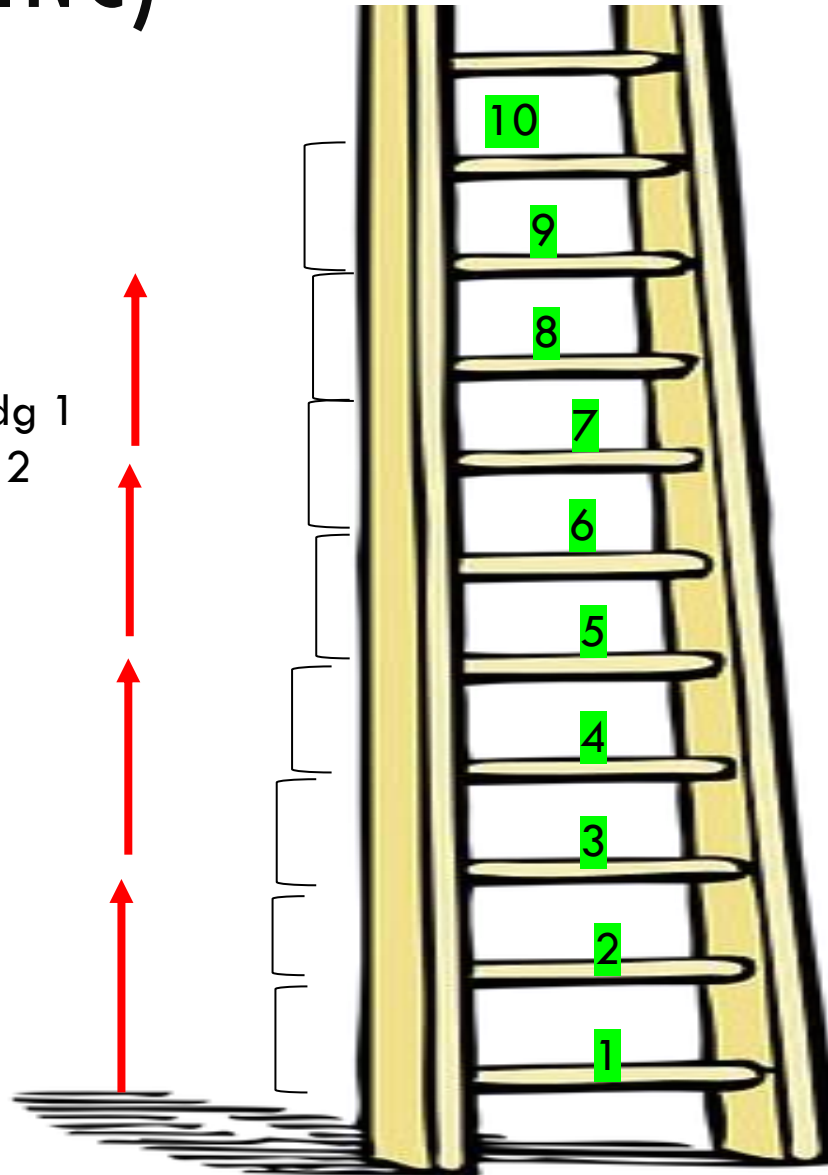
ADA SYARAT BERHENTI "KHUSUS"
TERGANTUNG KASUS

FOR ----→ WHILE

WHILE (belum tentu) FOR

INCREMENT (INC)

$\text{inc}++$ = lompatan bertambah dg 1
 $\text{inc}+=2$ lompatan bertambah dg 2

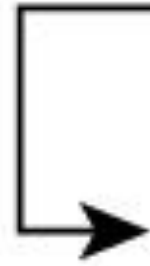


DECREMENT (DEC)

$\text{inc}--$ = lompatan berkurang dg 1
 $\text{inc}+=2$ tamganh dengan 2

BREAK

```
for var in sequence:  
    # codes inside for loop  
    if condition:  
        break
```



```
    # codes inside for loop  
# codes outside for loop
```

```
while test expression:  
    # codes inside while loop  
    if condition:  
        break
```



```
    # codes inside while loop  
# codes outside while loop
```

BREAK

```
number = 0

for number in range(10):
    number = number + 1

    if number == 5:
        break    # break here

    print('Number is ' + str(number))

print('Out of loop')
```

BREAK

```
break_examples.py ×  
9  
10     count = 10  
11  
12     while count > 0:  
13         print(count)  
14         if count == 5:  
15             break  
16         count -= 1  
17  
  
Run: break_examples ×  
/Users/pankaj/Documents/PycharmProjects/Pytho  
10  
9  
8  
7  
6  
5  
  
Process finished with exit code 0
```

CONTOH : F O R & W H I L E

Tulislah bilangan 1 sampai dengan
bilangan 5

<< 1, 2, 3, 4, 5>>

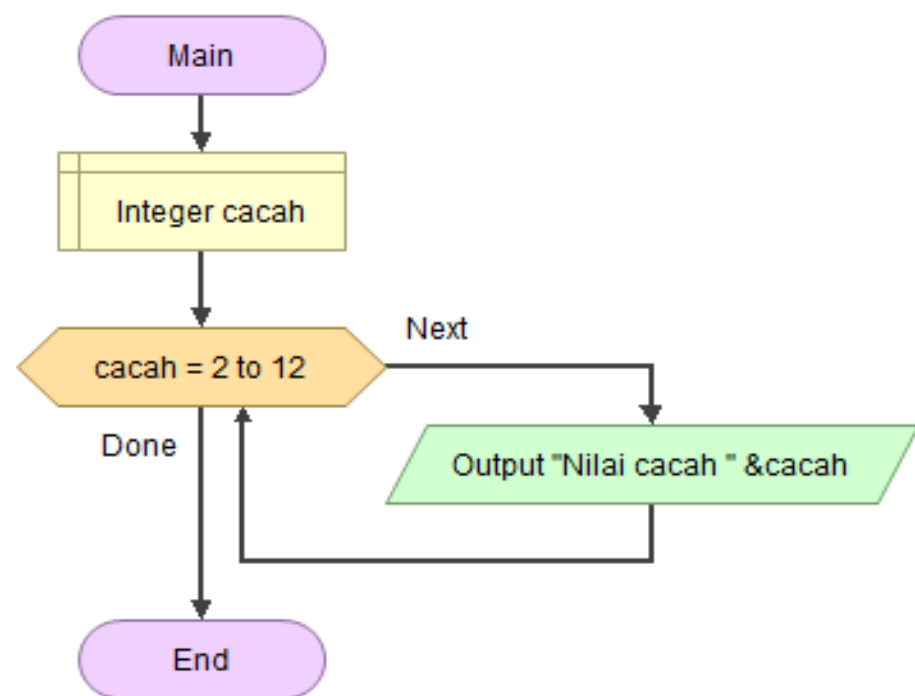
1. Berapa banyaknya
perulangan/looping-nya?
2. Jika counter awal = 1, berapa
nilai counter akhir-nya?

1

Tulislah bilangan genap
antara 10 sampai 20
<< 12, 14, 16, 18, 20>>

1. Berapa banyaknya
perulangan/looping-
nya?
2. Jika counter awal = 1,
berapa nilai counter
akhir-nya?

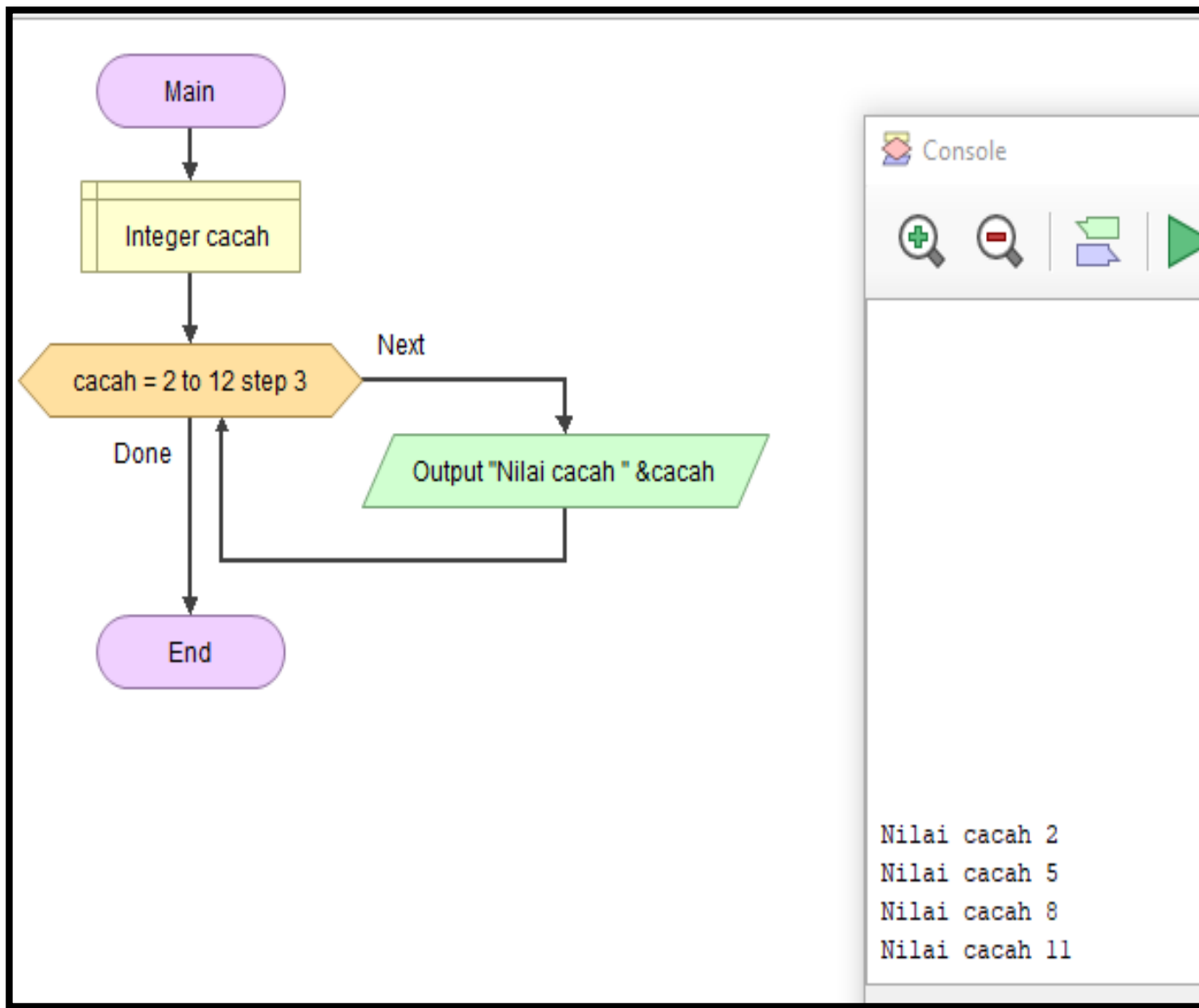
2



Console

```
Nilai cacah 2  
Nilai cacah 3  
Nilai cacah 4  
Nilai cacah 5  
Nilai cacah 6  
Nilai cacah 7  
Nilai cacah 8  
Nilai cacah 9  
Nilai cacah 10  
Nilai cacah 11  
Nilai cacah 12
```

The screenshot shows a console window with the output of the program, displaying the values of 'cacah' from 2 to 12, each preceded by the text 'Nilai cacah '.



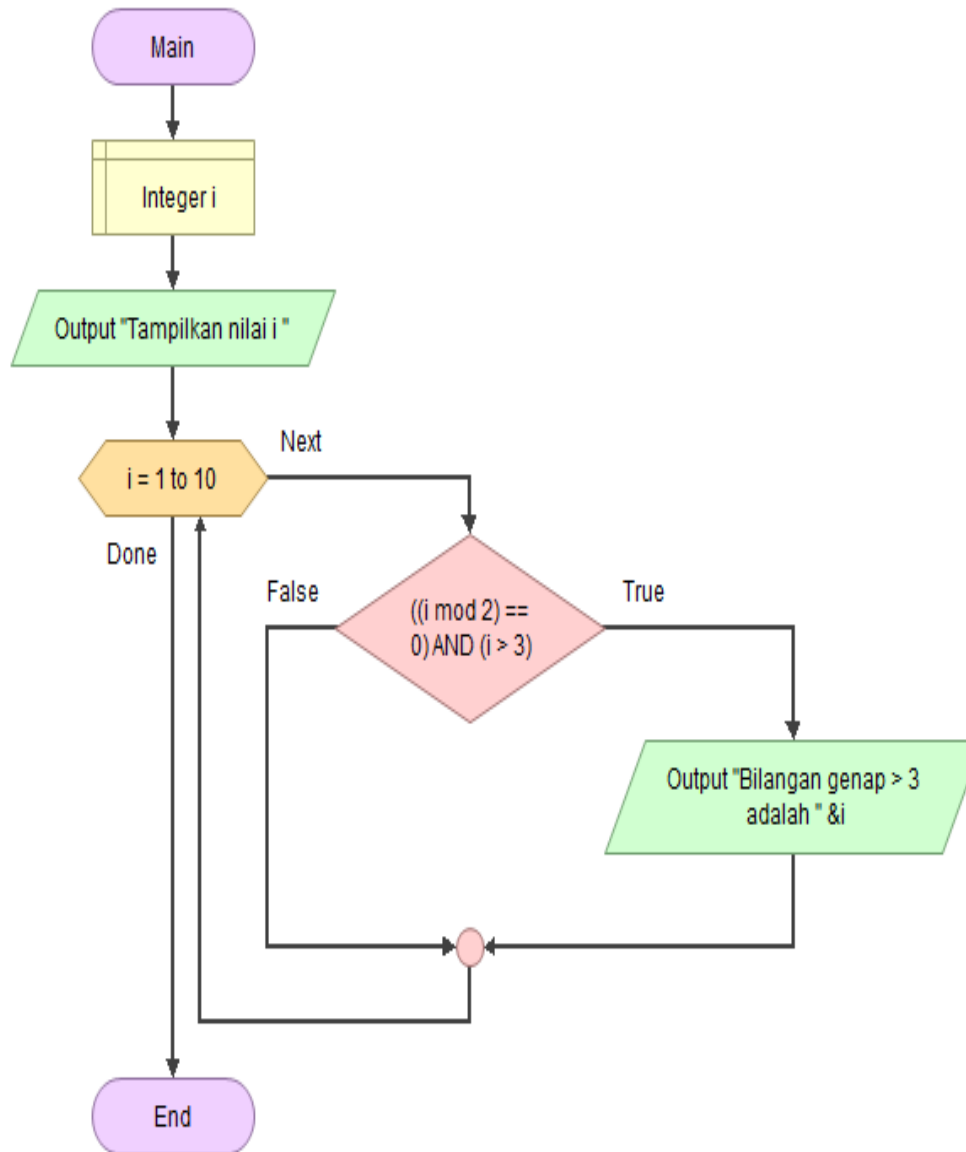
The Source Code Viewer window displays the following Python code:

```
for cacah in range(2, 12 + 3, 3):
    print("Nilai cacah = " + str(cacah))
```

Console

Tampilkan nilai i
Bilangan genap > 3 adalah 4
Bilangan genap > 3 adalah 6
Bilangan genap > 3 adalah 8
Bilangan genap > 3 adalah 10

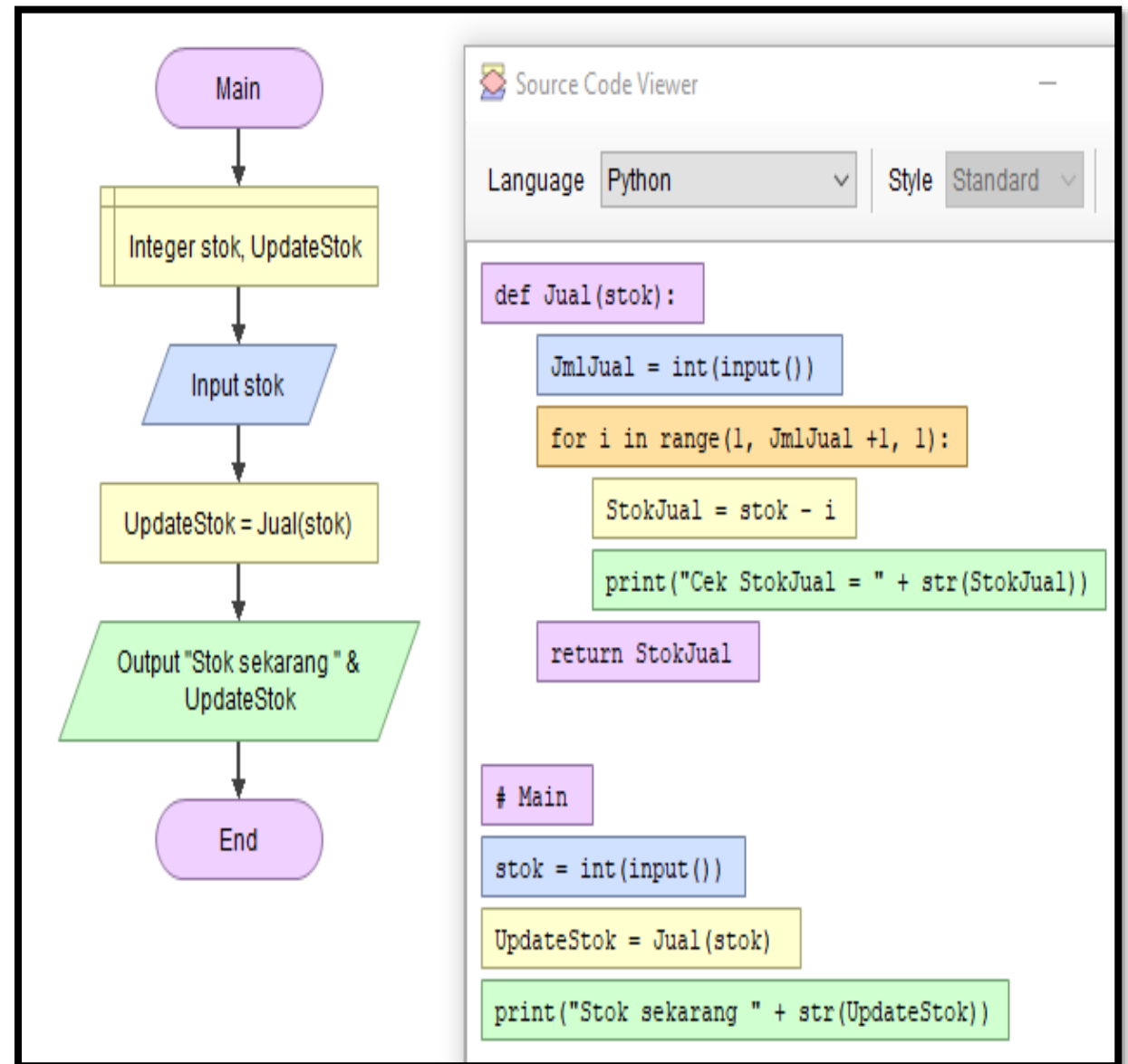
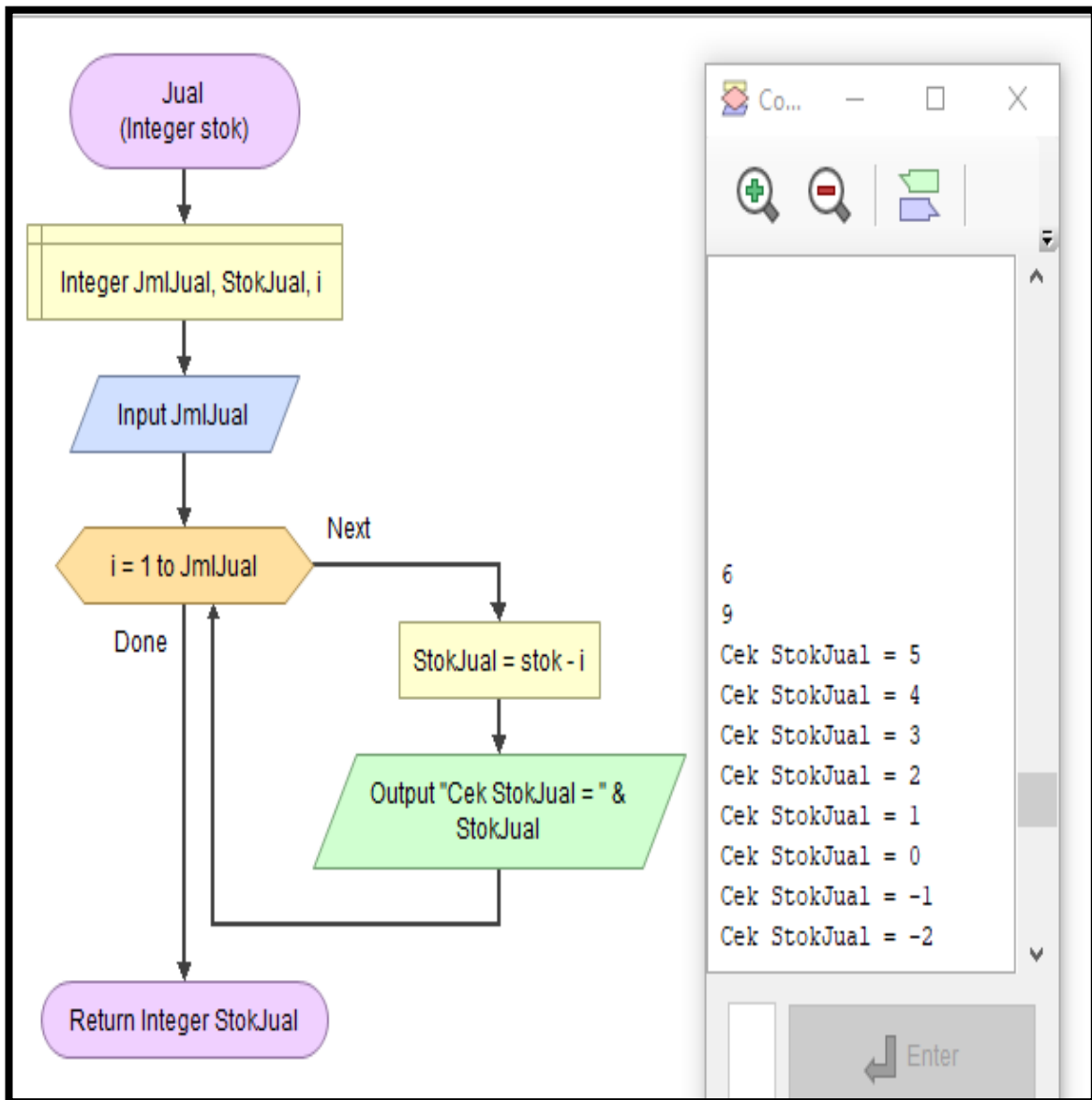
Enter



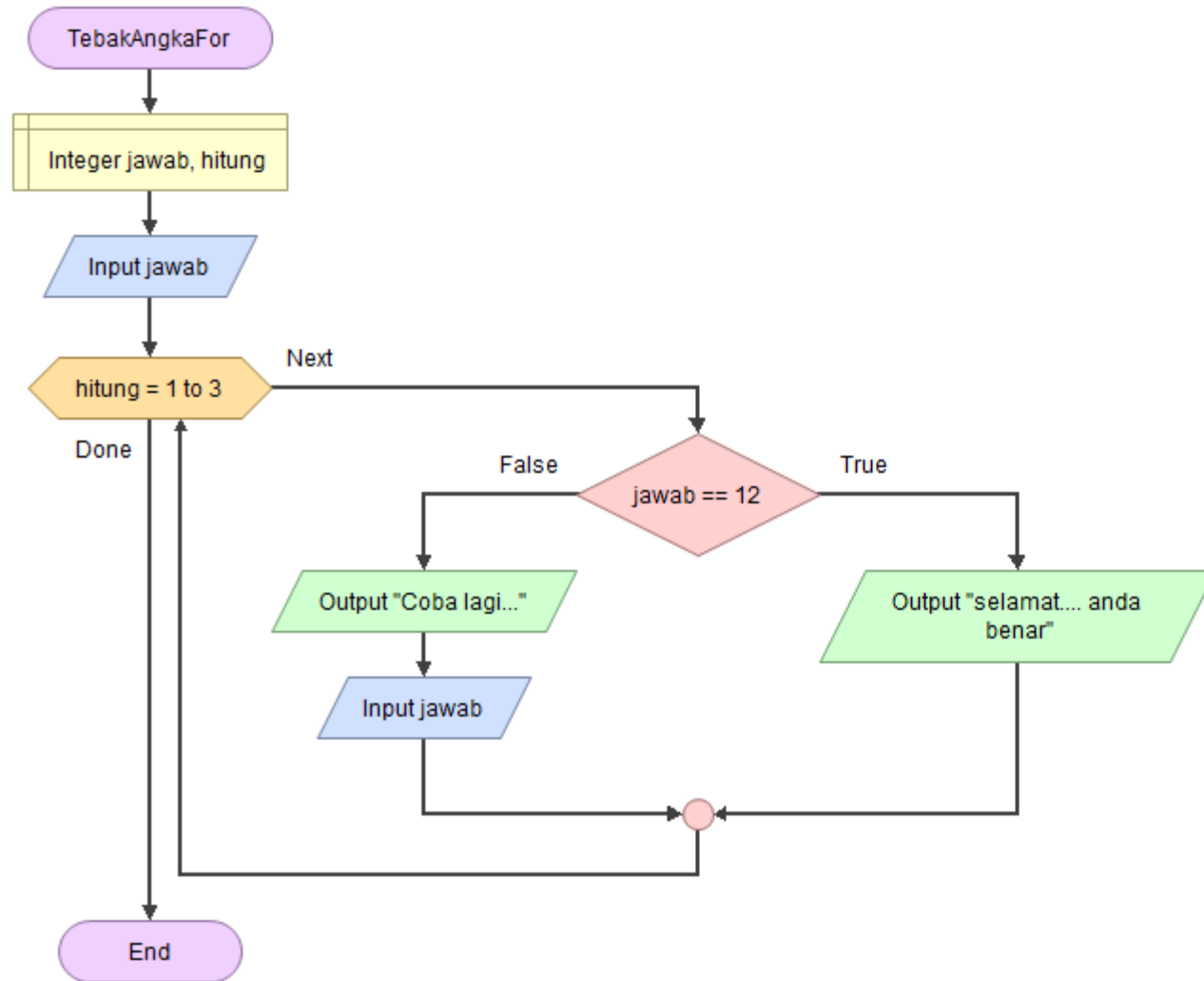
Source Code Viewer

Language Python Style Standard

```
print("Tampilkan nilai i ")
for i in range(1, 10 + 1, 1):
    if i % 2 == 0 and i > 3:
        print("Bilangan genap > 3 adalah " + str(i))
```



FOR



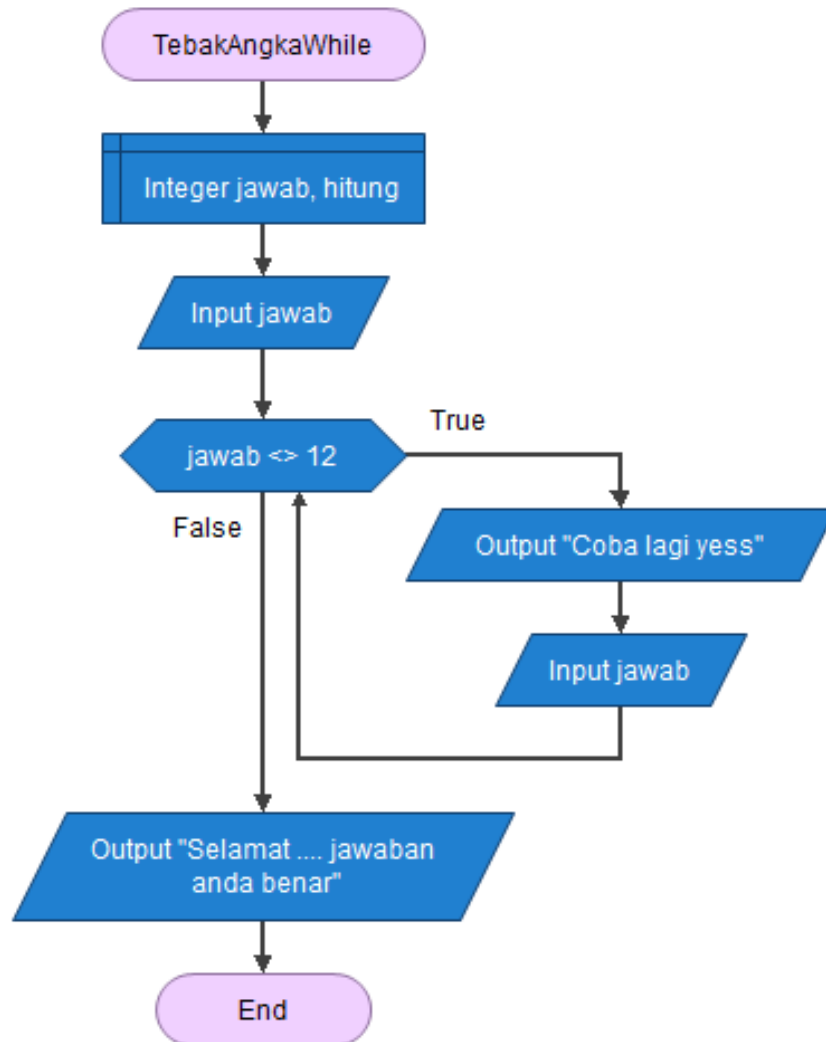
Console

```
Ini main() tebak angka
Berapakah 3 * 4 ?
3
Coba lagi...
3
Coba lagi...
5
Coba lagi...
6
```

The console window shows the program's output. It prompts the user to guess the result of 3 * 4. The user enters 3, 3, and 5, each time receiving the prompt 'Coba lagi...'. On the fourth attempt, the user enters 6, and the program outputs 'selamat.... anda benar'.

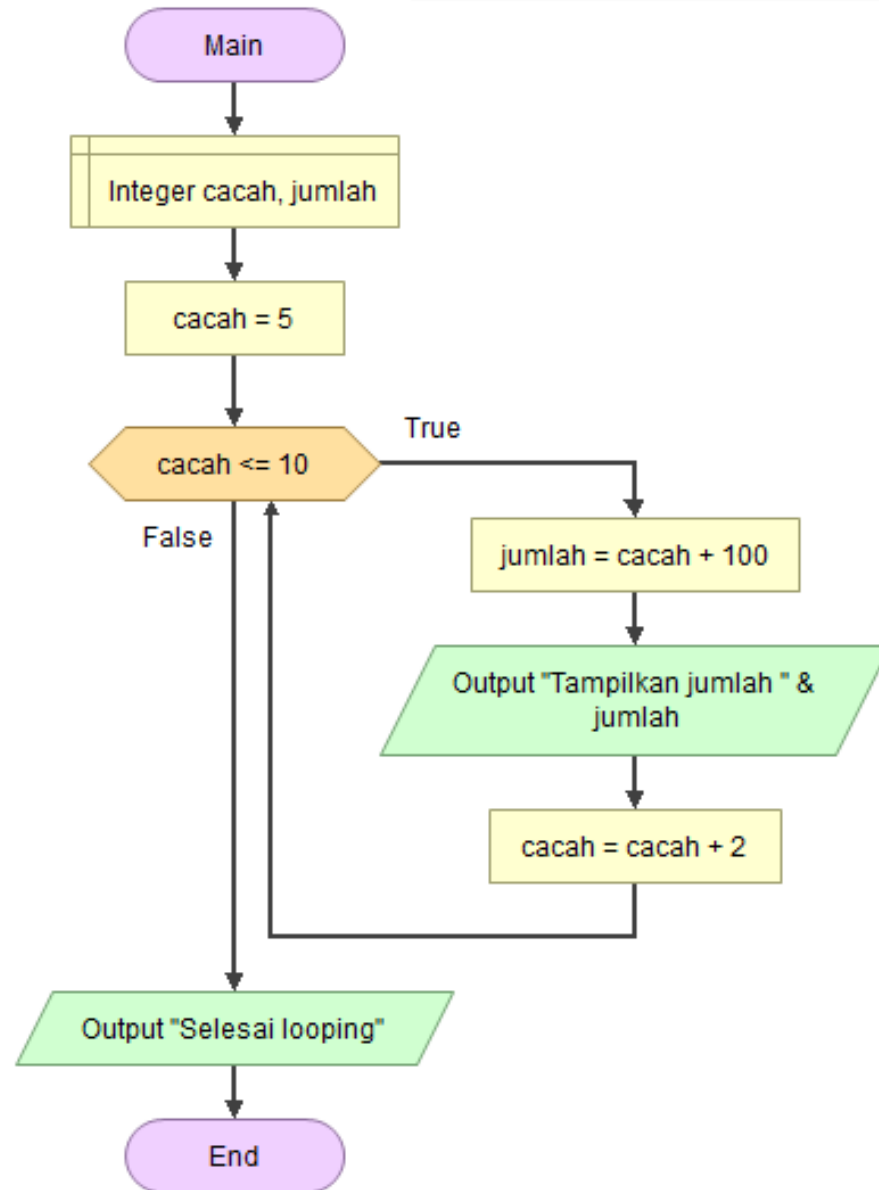
```
Ini main() tebak angka
Berapakah 3 * 4 ?
12
selamat.... anda benar
selamat.... anda benar
selamat.... anda benar
```


WHILE



```
Ini main() tebak angka
Berapakah 3 * 4 ?
2
Coba lagi yess
4
Coba lagi yess
5
Coba lagi yess
7
Coba lagi yess
8
Coba lagi yess
1
Coba lagi yess
5
Coba lagi yess
7
Coba lagi yess
12
```

WHILE

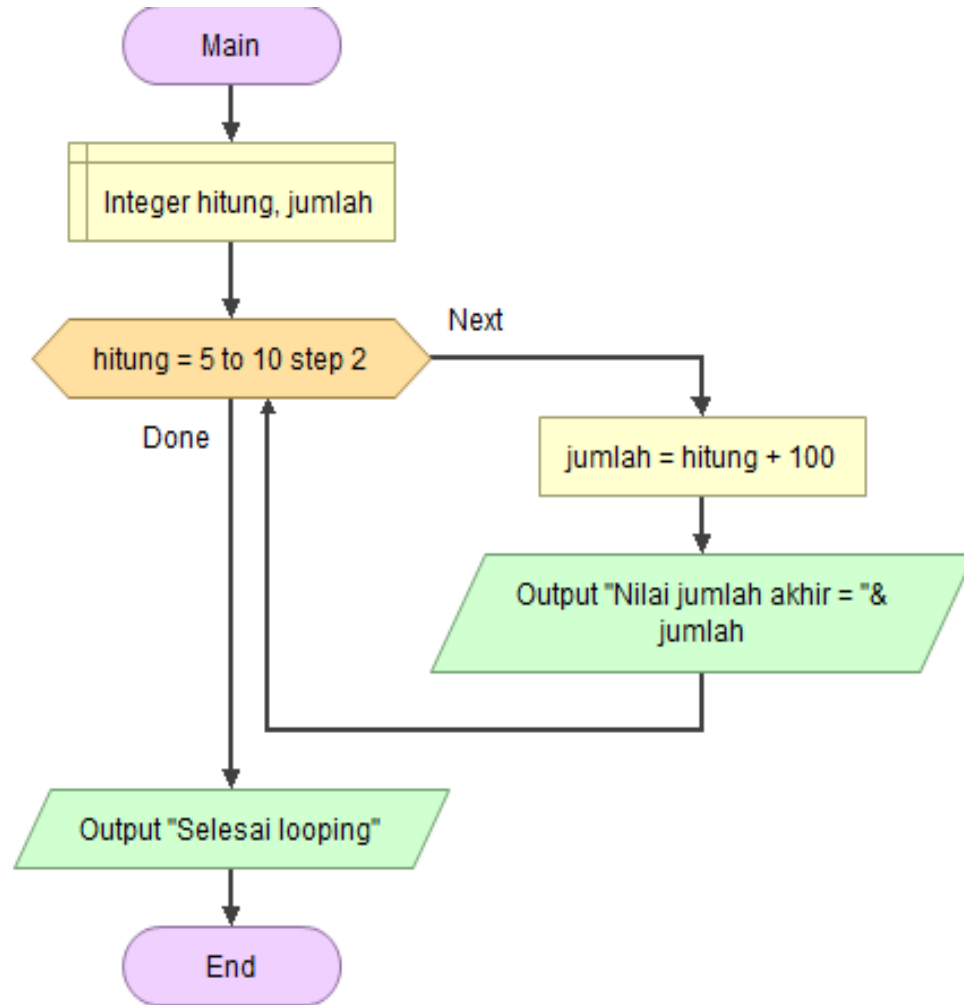


Tampilkan jumlah 105

Tampilkan jumlah 107

Tampilkan jumlah 109

FOR



Nilai jumlah akhir = 105

Nilai jumlah akhir = 107

Nilai jumlah akhir = 109

Selesai looping